

FAR-FUTURE CALENDAR ACCESS

Generalized Microsoft Graph discovery experiment for calendarView ranges in years 2040-2099.

HUNTING**CONFIDENCE: MEDIUM**

HOLLOWGRAPH / GENERALIZED HUNT

Measure far-future calendar access, identify approved applications and collect raw events for a tenant baseline.

GRAPH CLIENT**AUTHENTICATION****CALENDARVIEW GET****FAR-FUTURE RANGE****ANALYST REVIEW****REPORT DATE****22 July 2026****VERSION****v2.0****PRIMARY TELEMETRY****Microsoft Sentinel ·
MicrosoftGraphActivityLogs****ATT&CK / SCOPE****T1102.002 · T1071.001**

Independent defensive research. Source facts, analytical assessment, assumptions and detection logic are intentionally separated. Every query requires environment-specific validation.

THREAT RESEARCH EDITORIAL STANDARD · HTML MASTER V2

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EXECUTIVE ASSESSMENT

Decision-grade summary for threat research, detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Far-Future Calendar Access

This hunt generalizes the far-future date pattern observed in HOLLOWGRAPH and identifies calendarView requests whose startDateTime targets years 2040-2099.

WHY IT MATTERS

Operational detection value

The behavior is directly observable when RequestUri is retained, but maliciousness is not established. Approved applications may legitimately use wide or synthetic date ranges.

Medium 01

ANALYTICAL CONFIDENCE



Focused 02

TELEMETRY COVERAGE



Hunting 03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACT

Documented behavior

The source-observed implant queried calendarView for a date in 2050 through Microsoft Graph.

ANALYTICAL ASSESSMENT

Interpretation

Far-future calendar access is a useful discovery dimension, not a standalone malicious condition.

DETECTION ASSUMPTION

Required condition

RequestUri retains a readable startDateTime parameter and the regex covers the collected encoding format.

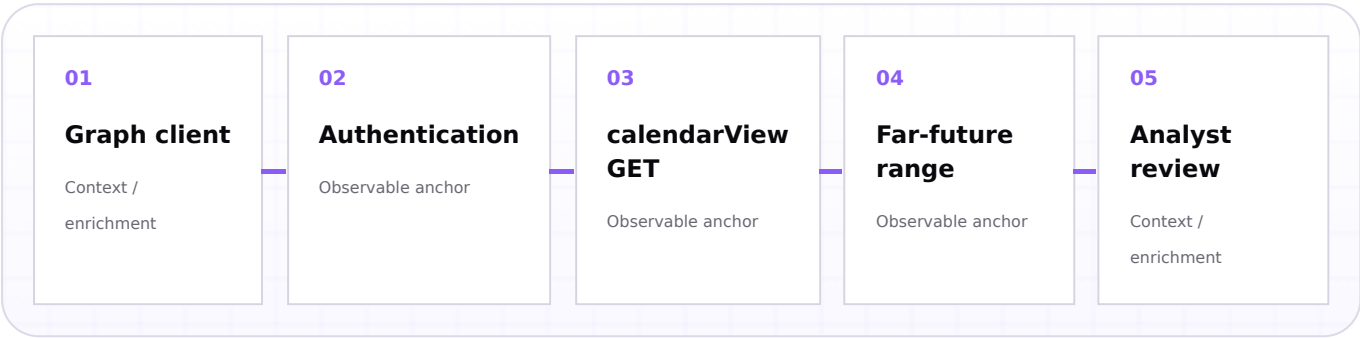
Evidence boundary

It does not prove HOLLOWGRAPH, covert tasking, malicious attachment access or compromised credentials.

THREAT MECHANICS

Behavioral decomposition and observable evidence boundaries.

ATTACK FLOW



STAGE-BY-STAGE DETECTION VALUE

STAGE	BEHAVIOR	EVIDENCE EXPECTATION	USE
01	Graph client	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
02	Authentication	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
03	calendarView GET	The source-observed implant queried calendarView for a date in 2050 through Microsoft Graph.	Primary anchor
04	Far-future range	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
05	Analyst review	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment

Observable core
A match indicates a calendarView request targeting a year from 2040 through 2099.

Non-observable boundary
It does not prove HOLLOWGRAPH, covert tasking, malicious attachment access or compromised credentials.

TELEMETRY & EVIDENCE MODEL

Minimum data contract required for a defensible implementation.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Sentinel · MicrosoftGraphActivityLogs

The primary platform is selected because it observes the strongest source-supported behavior with the most direct event relationship.

EVIDENCE BOUNDARY

A match is an investigative lead.

It does not prove HOLLOWGRAPH, covert tasking, malicious attachment access or compromised credentials.

REQUIRED FIELD CONTRACT

TIER	DATA	PURPOSE	RELIABILITY
REQUIRED	TimeGenerated, RequestMethod, RequestUri, ResponseStatusCode, AppId, ServicePrincipalId, UserId, IPAddress, UserAgent	Core filtering, grouping and investigation output.	Documented schema; population must be verified.
OPTIONAL	Signer, hash, prevalence, file origin, software inventory, device group.	Benign separation and prioritization.	Environment-dependent.
ENRICHMENT	Threat intelligence, change windows, asset criticality and identity context.	Disposition and escalation.	Integration-dependent.

Completeness

Confirm table coverage, retention, field population and event latency before measuring prevalence.

Normalization

Validate case, path, quoting, URI encoding and command-line preservation against raw events.

Visibility gap

The generalized date range can overlap legitimate software and must not be promoted without baseline evidence.

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DETECTION LOGIC

Source-grounded implementation with explicit assumptions and outcome boundaries.

HUNTING

DETECTION OBJECTIVE

Measure far-future calendar access, identify approved applications and collect raw events for a tenant baseline.

IMPLEMENTATION STATUS

Hunting - Untested implementation sketch

Syntax and schema are documented; local execution, field population and false-positive performance are not established.

KQL · DEMONSTRATIVE

```
// Title: Far-future Microsoft Graph calendarView access
// Status: Hunting query
// Confidence: High for telemetry observability; Medium for maliciousness
// Purpose: Identify calendarView requests whose startDateTime targets years
//          2040-2099, generalizing the far-future calendar behavior observed
//          in HOLLOWGRAPH.
// Required table: MicrosoftGraphActivityLogs
// Required fields: TimeGenerated, RequestMethod, RequestUri,
//                  ResponseStatusCode, AppId, ServicePrincipalId, UserId,
//                  IPAddress, UserAgent
// MITRE ATT&CK: T1102.002, T1071.001
//
// This query must not be promoted directly to a production rule. First
// establish which approved applications legitimately query far-future dates.
// The regex assumes RequestUri retains a readable startDateTime parameter.

MicrosoftGraphActivityLogs
| where TimeGenerated >= ago(30d)
| where RequestMethod == "GET"
| where RequestUri contains "/calendarView"
| where RequestUri matches regex @"(?!)[?&]startDateTime=20[4-9][0-9]-"
| project
  TimeGenerated,
  AppId,
  ServicePrincipalId,
  UserId,
  IPAddress,
  UserAgent,
  RequestUri,
  ResponseStatusCode
```

What a match means

A match indicates a calendarView request targeting a year from 2040 through 2099.

What a match does not prove

It does not prove HOLLOWGRAPH, covert tasking, malicious attachment access or compromised credentials.

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TRIAGE & INVESTIGATION

Evidence-led workflow for defensible disposition.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the match.
Review the raw event, timestamp, entity relationship and exact matching fields.
- 02

Establish legitimacy.
Identify the software or change owner, expected path, signer and authorized maintenance window.
- 03

Reconstruct origin.
Pivot to process, file, identity, network or application events before and after the match.
- 04

Measure prevalence.
Compare hashes, paths, commands, users and devices across the retained fleet.
- 05

Document confidence.
Separate confirmed evidence, inference, assumptions and unresolved gaps.

DECISION MODEL

Approved context → tune narrowly

Known owner, expected software path and repeatable behavior can support a documented exclusion.

Unknown context → deepen review

Rare or risky context without corroboration remains suspicious or insufficient evidence.

Corroborated activity → escalate

Malicious follow-on process, file, network, persistence or identity evidence supports incident escalation.

Host

Coverage, exposure, recent changes and neighboring alerts.

User

Session, intent, role and authorization context.

Process

Parent, child, path, command, signer and original filename.

File

Hash, signature, origin, prevalence and creation timeline.

Network

DNS, proxy, TLS and destinations following the event.

Identity

Sign-ins, token use, privilege and resource access.

OUTCOME	EVIDENCE PROFILE	ACTION
Benign	Approved and repeatable behavior with strong ownership.	Document and tune narrowly.
Suspicious	Risky context without decisive malicious corroboration.	Deepen endpoint or application review.
Likely malicious	Source-compatible match plus corroborated malicious activity.	Escalate incident response.
Insufficient evidence	Missing telemetry or unresolved ownership.	Retain evidence and validate gaps.

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FALSE POSITIVES & VALIDATION

Operational controls required before alerting.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	LIKELIHOOD	TUNING ACTION
Calendar migration tools	High	Review owner, path, signer, prevalence and change context before exclusion.
Synthetic testing	High	Review owner, path, signer, prevalence and change context before exclusion.
Archival systems	Med	Review owner, path, signer, prevalence and change context before exclusion.
Application date-boundary validation	Med	Review owner, path, signer, prevalence and change context before exclusion.

Baseline plan

- Run retrospectively over a representative retention window.
- Measure volume by device, user, process, path and software owner.
- Inspect recurring legitimate patterns before adding exclusions.
- Preserve raw examples for peer review.

Validation procedure

- Confirm every required field against raw events.
- Execute only safe, authorized lab or historical validation.
- Compare query output with endpoint or application ground truth.
- Record success, rejection and sensor-failure criteria.

PROMOTION GATE

Hunting

Source-grounded hypothesis.

Candidate

Schema and baseline complete.

Production

Measured precision and runbook.

Optimized

Stable feedback loop.

Promotion criteria

Known alert volume, documented exclusions, representative validation, triage ownership, data-quality monitoring and measurable precision.

Failure condition

The generalized date range can overlap legitimate software and must not be promoted without baseline evidence.

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ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

COVERAGE MAPPING

MAPPING	COVERAGE STATEMENT	BOUNDARY
T1102.002 · T1071.001	Partial behavioral or campaign-artifact coverage of the source-supported observable.	It does not prove HOLLOWGRAPH, covert tasking, malicious attachment access or compromised credentials.

ANALYST CONCLUSION

Assessment, not certainty

The query is assessed as a medium-confidence hunting experiment for establishing prevalence and legitimate application ownership.

Residual risk

The generalized date range can overlap legitimate software and must not be promoted without baseline evidence.

REFERENCES

- 1. Group-IB - HOLLOWGRAPH: Turning Microsoft Graph into a Covert C2 Channel: [https://www\[group-ib\].com/blog/hollowgraph-microsoft-365/](https://www[group-ib].com/blog/hollowgraph-microsoft-365/)
- 2. Microsoft Learn - Microsoft Graph activity logs overview: <https://learn.microsoft.com/en-us/graph/microsoft-graph-activity-logs-overview>
- 3. Microsoft Learn - MicrosoftGraphActivityLogs table reference: <https://learn.microsoft.com/en-us/azure/azure-monitor/reference/tables/microsoftgraphactivitylogs>
- 4. T1102.002 - Web Service: Bidirectional Communication: <https://attack.mitre.org/techniques/T1102/002/>
- 5. T1071.001 - Application Layer Protocol: Web Protocols: <https://attack.mitre.org/techniques/T1071/001/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v2.0	22 July 2026	Hunting	Detection Engineering

Change summary

Migrated to the permanent Threat Research Editorial HTML master; technical query and evidence classification preserved.

Publication control

No customer data, credentials or environment-specific internal indicators are included. URLs are defanged in the publication artefact.